Limiting Number of States + Non-Linear states in synapse

Jaskirat Singh Maskeen

B. Tech. ’23 CSE

Indian Institute of Technology Gandhinagar

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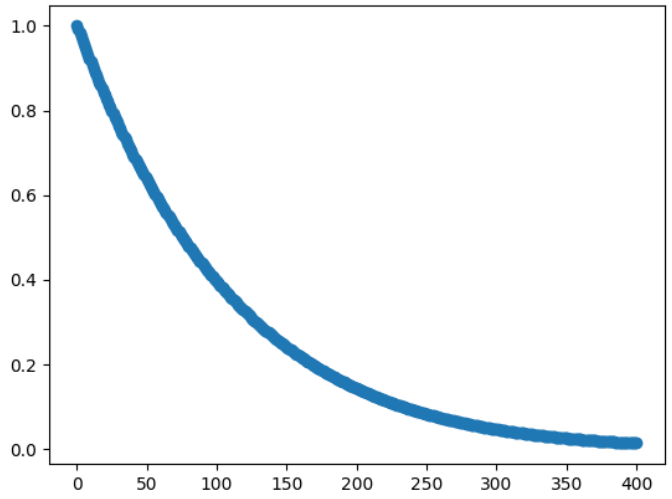
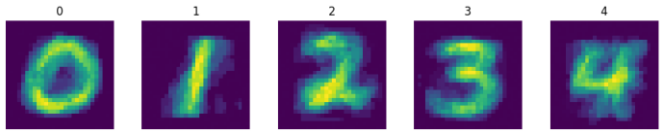
Continuous weight levels

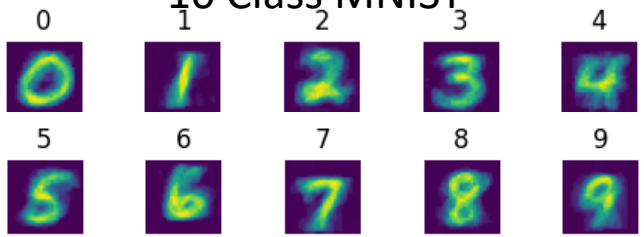
weight change vs. number of pulses to synapse



5 Class MNIST



10 Class MNIST

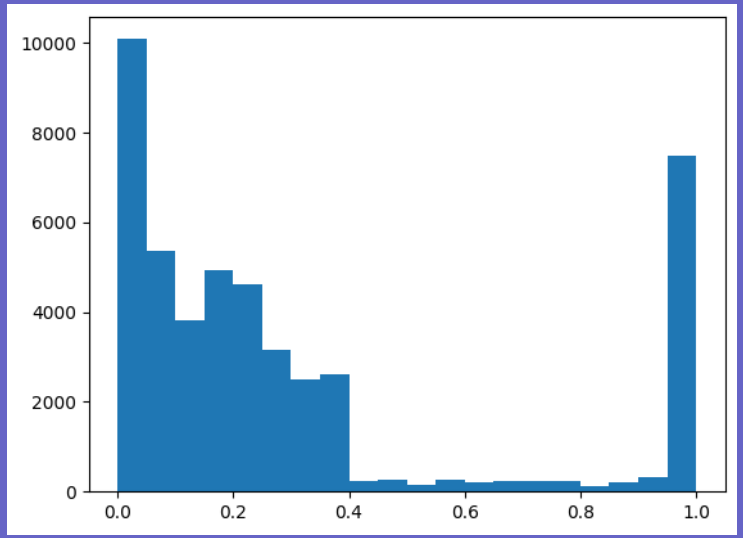


5 Class MNIST

Training Images: 100
Testing Images: 1500
Architecture: 784 x 60

Accuracy: 91.33%

Weight distribution:
Number of synapse vs weight

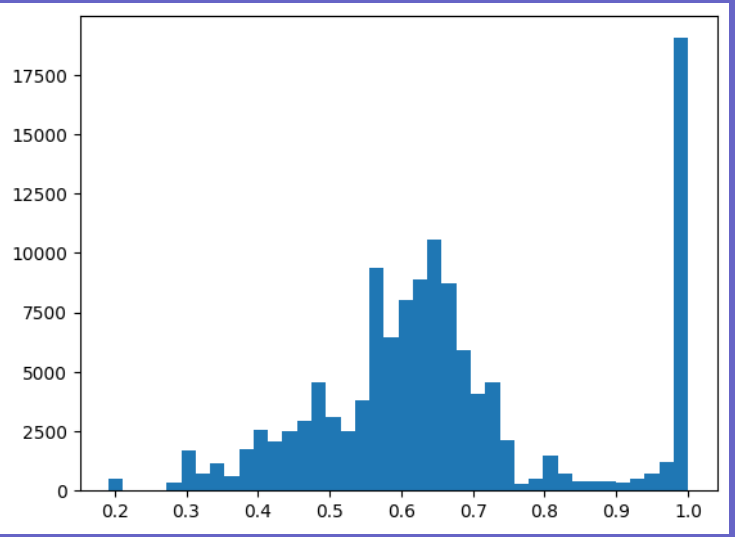


10 Class MNIST

Training Images: 200
Testing Images: 1500
Architecture: 784 x 160

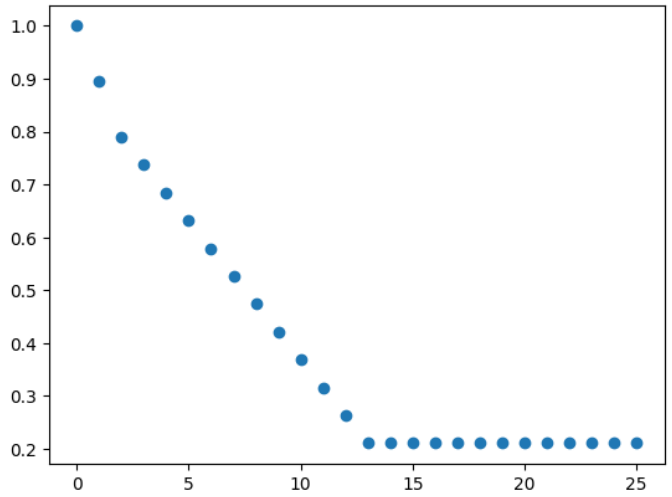
Accuracy: 74%

Weight distribution:
Number of synapse vs weight

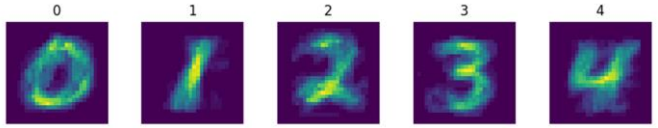


Uniform weight levels

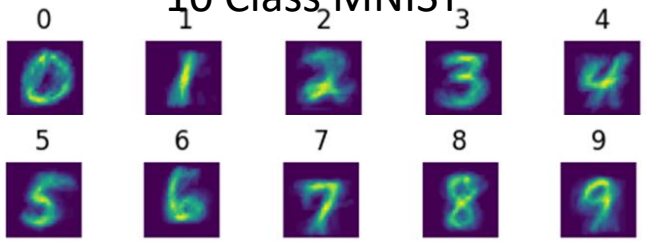
weight change vs. number of pulses to synapse



5 Class MNIST



10 Class MNIST

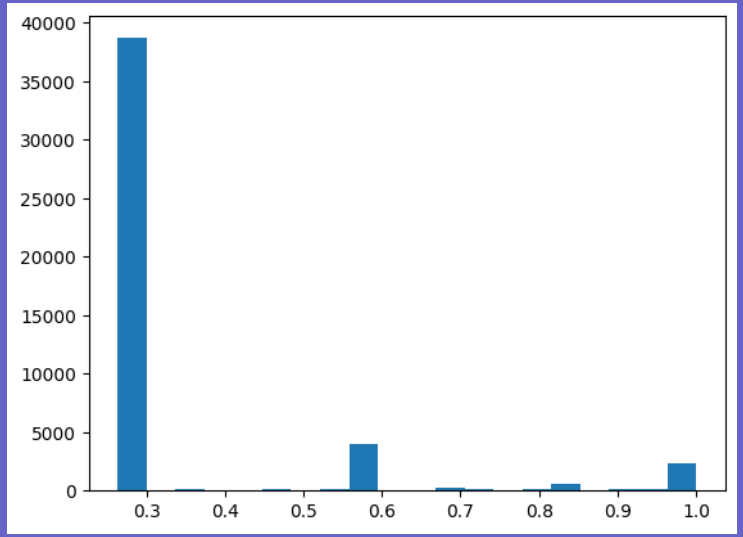


5 Class MNIST

Training Images: 100
Testing Images: 1500
Architecture: 784 x 60

Accuracy: 88.46%

Weight distribution:
Number of synapse vs weight



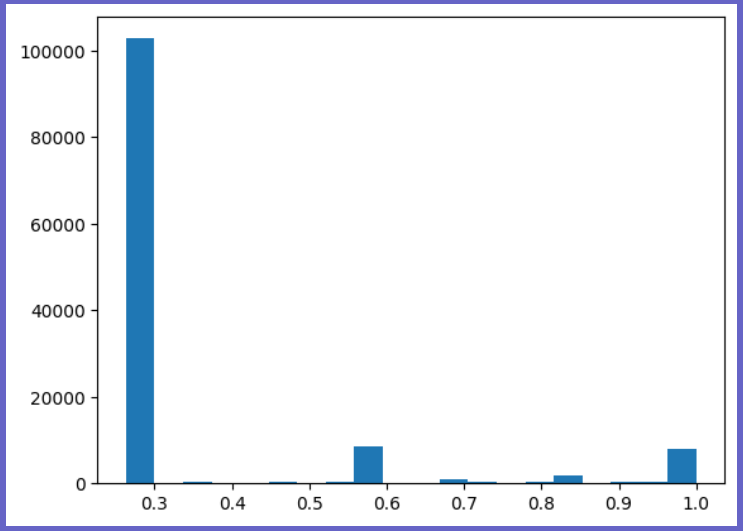
10 Class MNIST

Training Images: 200
Testing Images: 1500
Architecture: 784 x 160

Accuracy: 71%

Weight distribution:
Number of synapse vs weight

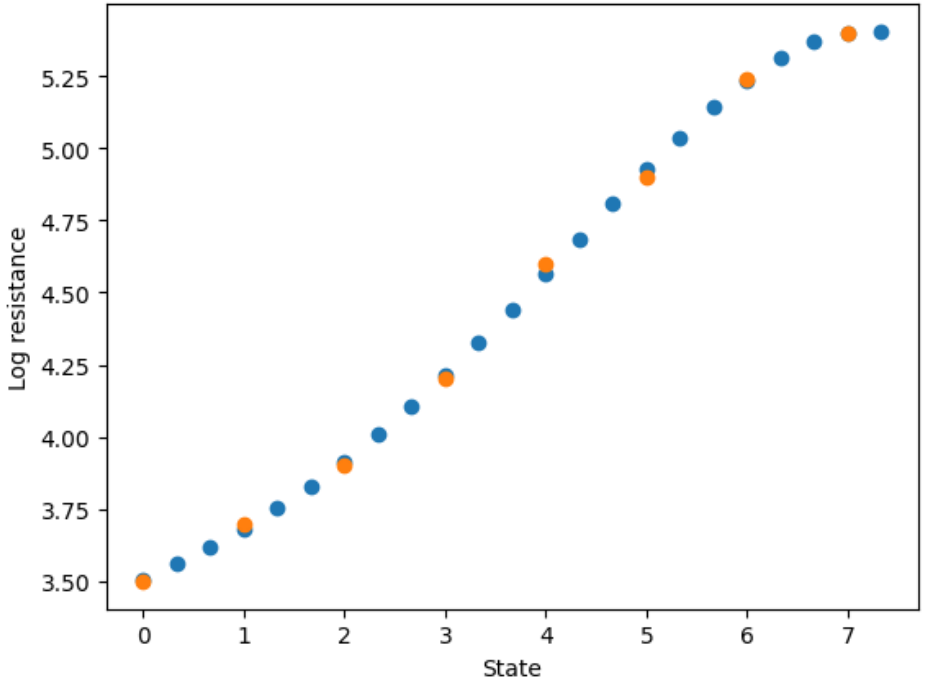
Wrong Training Example

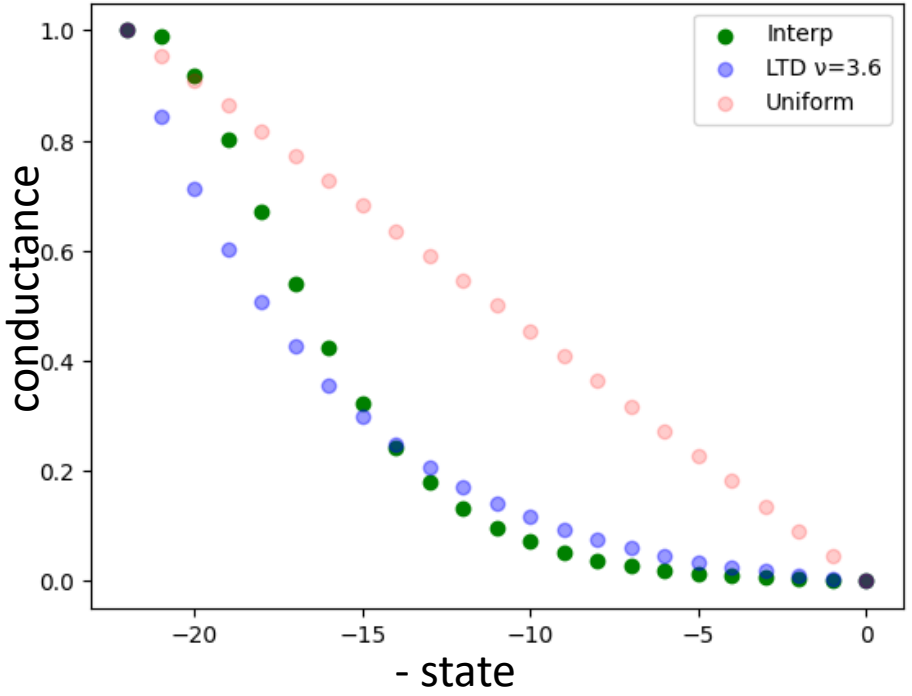


Experimental RRAM data

- The orange points are from the data provided by Harsh bhaiya.
- The blue points are the interpolated ***n*** points (a cubic interpolation)
- I have fitted a mathematical function, which I read in a paper

$$w\ (LTD) = \ normalized\ g = g_{\max} - \alpha\left(1 - e^{-\nu\left(1-\frac{p}{256}\right)}\right)$$



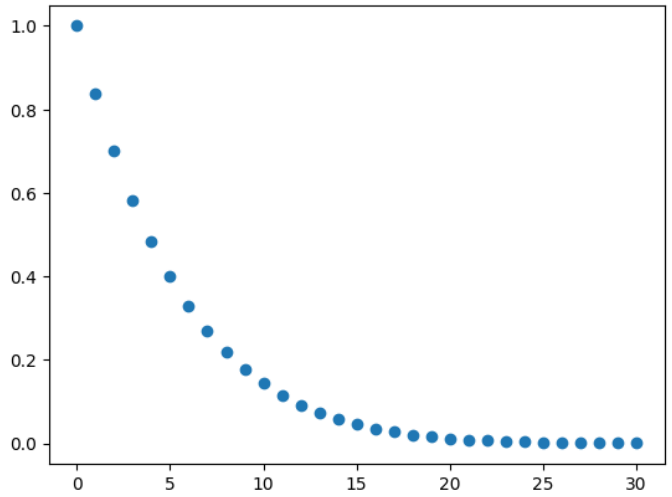


$$\alpha = \frac{g_{\max} - g_{\min}}{1 - e^{-\nu}}$$

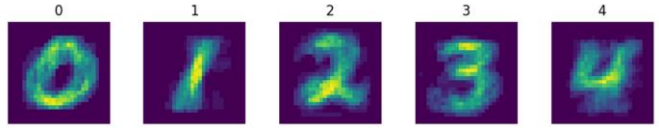
https://doi.org/10.3389/fncom.2021.646125

Non-Linear weight levels

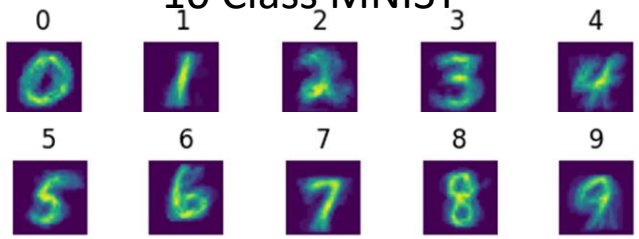
weight change vs. number of pulses to synapse



5 Class MNIST



10 Class MNIST

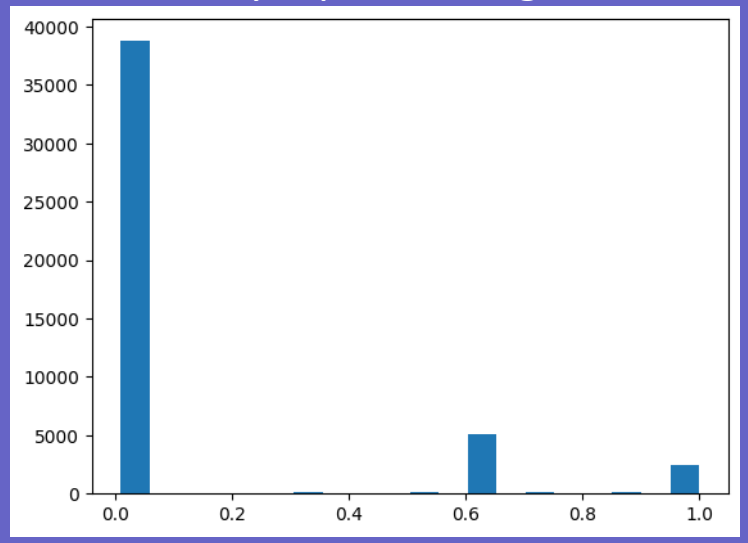


5 Class MNIST
25 Levels

Training Images: 100
Testing Images: 1500
Architecture: 784 x 60

Accuracy: 91%

Weight distribution:
Number of synapse vs weight

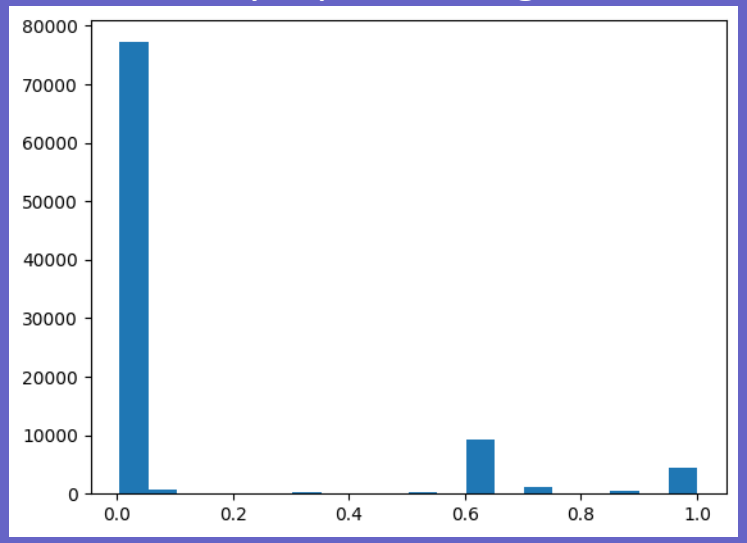


10 Class MNIST
25 Levels

Training Images: 200
Testing Images: 1500
Architecture: 784 x 120

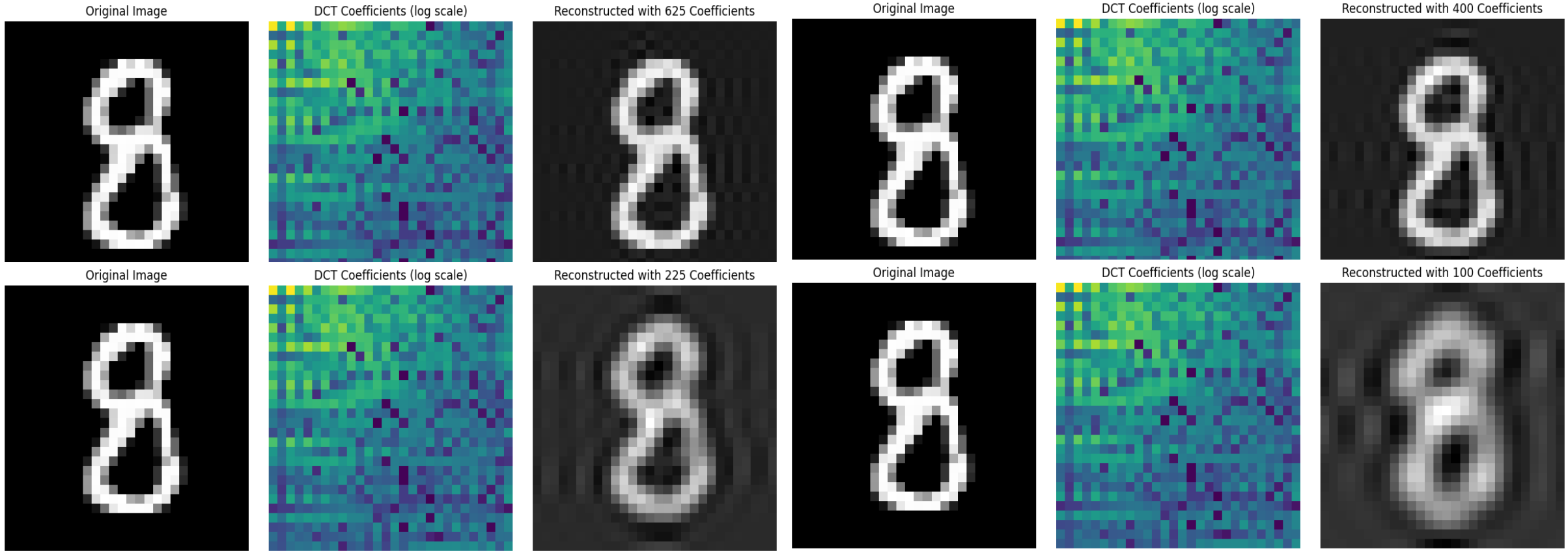
Accuracy: 71.2%

Weight distribution:
Number of synapse vs weight



Reducing Input Neurons

- Take DCT twice of the image (array)
- Take first k components (low frequency)
- Take inverse DCT twice to reconstruct image

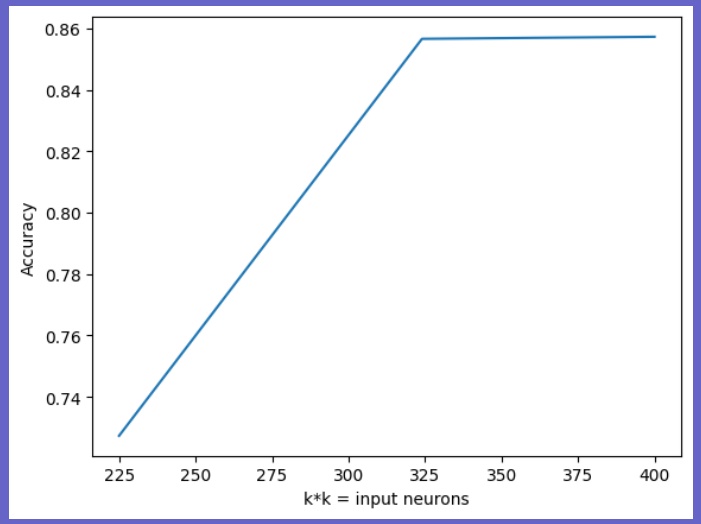


Reducing Input Neurons

- Accuracy on 5 class MNIST vs number of DCT coefficients

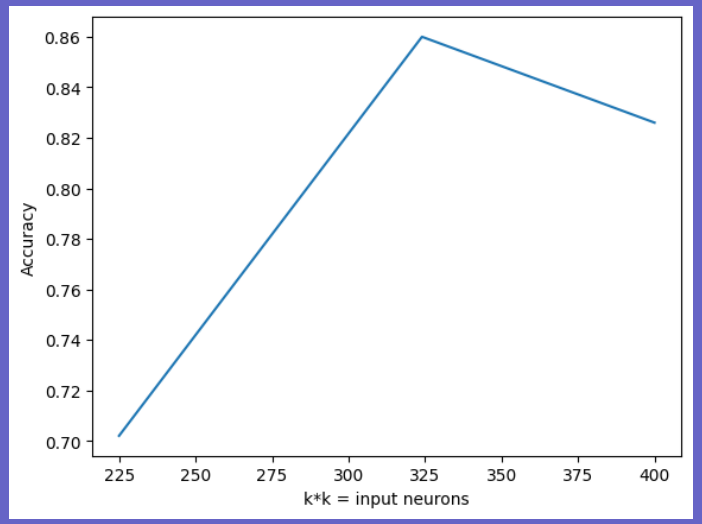
Continuous weights

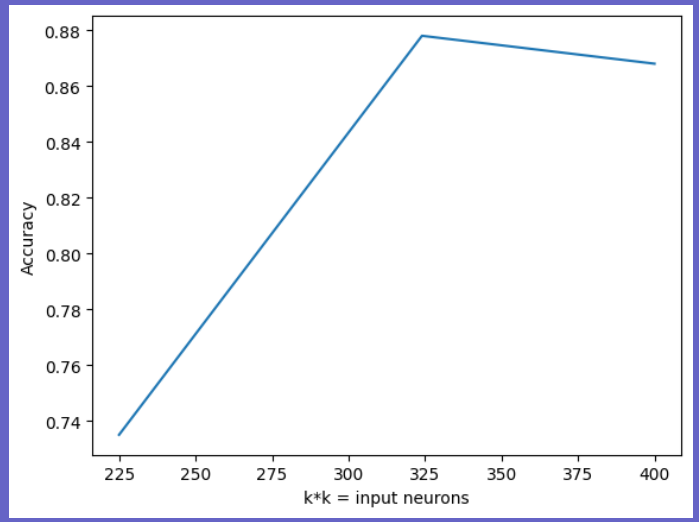
k=15 acc = 0.727
k=18 acc = 0.856
k=20 acc = 0.857



Uniform weights
25 levels

k=15 acc = 0.702
k=18 acc = 0.860
k=20 acc = 0.826



Non uniform weights
25 levels

k=15 acc = 0.727
k=18 acc = 0.856
k=20 acc = 0.857